

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 6 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTCVPE604A - INDUSTRIAL WASTE TREATMENT

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 The first step in minimizing the effects of industrial wastes is to reduce the: _____
 - A. Temperature
 - B. pH level
 - C. Volume
 - D. Odor
- 2 Equalization tanks in industrial wastewater treatment are used to:
 - A. Increase pH
 - B. Reduce BOD
 - C. Balance flow and pollutant concentration
 - D. Remove oil and grease
- 3 What is the primary objective of industrial waste treatment?
 - A. Reduce labour costs
 - B. Minimize environmental impact
 - C. Increase production efficiency
 - D. Improve product quality
- 4 What does the term 'zero liquid discharge' mean in industrial waste treatment?
 - A. Discharging treated water to rivers
 - B. Zero gas emissions
 - C. Recycling all wastewater within the plant and no liquid discharge
 - D. No solid waste generate
- 5 Biological treatment of industrial waste is ineffective when wastewater contains:
 - A. Nutrients
 - B. Toxic substances
 - C. Organic matter
 - D. Microorganisms

- 6 What is the primary function of the selective membrane in a reverse osmosis (RO) system?
 - A. To dissolve salts in wastewater
 - B. To allow only large molecules to pass
 - C. To neutralize acidic waste
 - D. To allow only water molecules to pass
- 7 If a water body has a WQI value of 96, it is categorized as:_____
 - A. Excellent
 - B. Good
 - C. Poor
 - D. Very Poor
- 8 Which industries are typically connected to CETPs?
 - A. IT companies
 - B. Small and medium-scale industries
 - C. Construction companies
 - D. Agricultural fields
- 9 The main objective of industrial waste treatment is to:
 - A. Recover water only
 - B. Protect public health and environment
 - C. Increase industrial production
 - D. Reduce operating cost only
- 10 Physical adsorption involves:_____
 - A. Van der Waals forces and multilayer formation
 - B. High specificity and irreversible layers
 - C. Chemical bonding
 - D. Use of acids and bases for reaction
- 11 What is the molecular weight of benzene (C_6H_6) used for ThOD calculations?
 - A. 70
 - B. 80
 - C. 92
 - D. 78
- 12 BOD of distillery spentwash is in the range of _____mg/Lit
 - a. 10,000 to 20,000
 - b. 20,000 to 40,000
 - c. 80,000 to 1,00,000
 - d. 1,50,000 to 2,00,000

Q2. Solve the following.

- A) Define : i. Grab Samples ii. composite sample 6
 For a industrial Wastewater five grab samples were taken.
 Readings of analysis (mg/L) are as follows:
 22, 18, 25, 20, 15
 Find: (i) Average, (ii) Peak, (iii) Range
- B) Explain the necessity of equalization and proportioning for industrial waste water 6
 treatment. Explain two types of Equalization tank

Q3. Solve the following.

- A) Enlist Neutralization methods for alkaline waste. Explain any two methods in 6
 detail.

- B) Enlist different waste volume reduction techniques. Explain any one in detail. 6
- Q4. Solve Any Two of the following.
- A) Reproduce a neat flow sheet for manufacturing of sugar industry . Also list sources of wastewater generation. 6
- B) Discuss treatment alternatives for Tannery waste. Draw Flowsheet/flowsheets 6
- C) Discuss in detail 'Massive Lime treatment' for paper and pulp industry 6
- Q5. Solve Any Two of the following.
- A) What is CETP. Discuss significance and types of CETP. 6
- B) Explain any one bioremediation technique 6
- C) Discuss salient features of Environmental Protection act 6
- Q6. Solve Any Two of the following.
- A) Compute WQI and classify water
Given (Drinking water sample): 6
- | Parameter | Observed | Standard | Ideal |
|-----------------|----------|----------|-------|
| pH | 7.6 | 8.5 | 7.0 |
| TDS (mg/L) | 450 | 500 | 0 |
| Chloride (mg/L) | 140 | 250 | 0 |
| Nitrate (mg/L) | 20 | 45 | 0 |
- B) Explain the concept of ISO 14000. Compare ISO 9000 and ISO 14000 6
- C) Enlist WQI methods and explain any one in detail. 6

*** End ***